

The Closed Interval Method To find the *absolute* maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b) .
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

EXAMPLE 8 Find the absolute maximum and minimum values of the function

$$f(x) = x^3 - 3x^2 + 1 \quad -\frac{1}{2} \leq x \leq 4$$

SOLUTION Since f is continuous on $[-\frac{1}{2}, 4]$, we can use the Closed Interval Method.

In Example 7(a) we found the critical numbers $x = 0$ and $x = 2$. Notice that each of these critical numbers lies in the interval $(-\frac{1}{2}, 4)$. The values of f at these critical numbers are

$$f(0) = 1 \quad f(2) = -3$$

The values of f at the endpoints of the interval are

$$f(-\frac{1}{2}) = \frac{1}{8} \quad f(4) = 17$$

Comparing these four numbers, we see that the absolute maximum value is $f(4) = 17$ and the absolute minimum value is $f(2) = -3$.

(4, 17) In this example the absolute maximum occurs at an endpoint, whereas the absolute minimum occurs at a critical number. The graph of f is sketched in Figure 15. ■

With graphing software or a graphing calculator it is possible to estimate maximum and minimum values very easily. But, as the next example shows, calculus is needed to find the *exact* values.

51–66 Find the absolute maximum and absolute minimum values of f on the given interval. **Question#01 from 51–56**

51. $f(x) = 12 + 4x - x^2$, $[0, 5]$

52. $f(x) = 5 + 54x - 2x^3$, $[0, 4]$

53. $f(x) = 2x^3 - 3x^2 - 12x + 1$, $[-2, 3]$

54. $f(x) = x^3 - 6x^2 + 5$, $[-3, 5]$

55. $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$, $[-2, 3]$

56. $f(t) = (t^2 - 4)^3$, $[-2, 3]$

the sign of $f'(x)$ changes from positive to negative at c . This is the following test.

The First Derivative Test Suppose that c is a critical number of a continuous function f .

- (a) If f' changes from positive to negative at c , then f has a local maximum at c .
- (b) If f' changes from negative to positive at c , then f has a local minimum at c .
- (c) If f' is positive to the left and right of c , or negative to the left and right of c , then f has no local maximum or minimum at c .

maximum and minimum values. It is a consequence of the Concavity Test, and it serves as an alternative to the First Derivative Test.

The Second Derivative Test Suppose f'' is continuous near c .

- (a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- (b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

29–30 Find the local maximum and minimum values of f using both the First and Second Derivative Tests. Which method do you prefer?

Question#01 from 29–30

29. $f(x) = 1 + 3x^2 - 2x^3$

30. $f(x) = \frac{x^2}{x - 1}$

8-70 Find the limit. Use l'Hospital's Rule. If there is a more elementary method, consider it. If l'Hospital's Rule doesn't apply, explain why.

8. $\lim_{x \rightarrow 3} \frac{x - 3}{x^2 - 9}$

Question#3 All

9. $\lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{x - 4}$

11. $\lim_{x \rightarrow 1} \frac{x^7 - 1}{x^3 - 1}$

13. $\lim_{x \rightarrow \pi/4} \frac{\sin x - \cos x}{\tan x - 1}$

15. $\lim_{t \rightarrow 0} \frac{e^{2t} - 1}{\sin t}$

17. $\lim_{x \rightarrow 1} \frac{\sin(x - 1)}{x^3 + x - 2}$

19. $\lim_{x \rightarrow \infty} \frac{\sqrt{x}}{1 + e^x}$

$\ln x$